WHITEBOX: A Topological Framework for Developmental Interpretability of Neural Network Learning

Arjun Paradkar • University of Arkansas • United States • Track: Natural & Applied Science (NAS)

1. RESEARCH QUESTION

The black box names seven distinct failures: we cannot see (P1) what a model learned, (P3) when bias formed, (P4) what it does not know, (P5) where it will fail, (P6) whether it understands or memorizes, or (P7) how its representations evolved. **Five had no structural solution.**

Four researchable questions:

- **RQ1.** Detect memorization vs. generalization *without* a held-out validation set?
- **RQ2.** Quantify how *robustly* each concept was learned?
- **RQ3.** Flag failure-prone inputs from internal geometry *before* output?
- **RQ4.** Locate and remove bias at the representation level *during* training?

Significance. Interpretability failure is the central barrier to trustworthy AI in medicine, hiring, and safety-critical systems. Current tools are post-hoc and output-level; none read the *learning process* structurally.

2. METHODOLOGY / PROJECT DESIGN

Core idea: apply persistent homology to activation point clouds across training epochs → a *topological film* $\mathcal{F}^l = (D_k^l(t))_{t=0\ldots T}$.

Pipeline: probe activations → PCA (15–30 dim) → Vietoris–Rips filtration → Ripser (maxdim 1) → persistence diagrams.

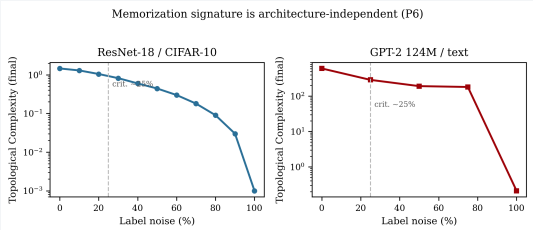
Three metrics:

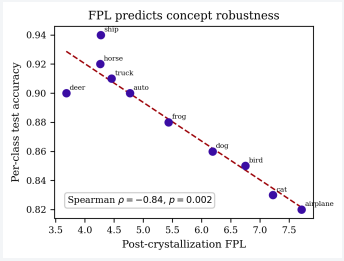
- **SV** $= W_2(D(t), D(t-1))$: rate of structural change
- **TC** $= \sum(d_i - b_i)^2$: amount of real structure
- **FPL**: epochs a concept's loops persist

Models / data (all public): ResNet-18 / CIFAR-10; GPT-2 124M; tiny ViT & TinyGPT (from scratch); WinoBias, COMPAS, Davidson corpora.

Validation: multi-seed, permutation tests, Mann–Whitney, cross-architecture replication, control vs. intervened runs. Hardware: single RTX 5070 (8GB).

3. DATA ANALYSIS & RESULTS





P6: TC separates memorize vs. generalize ∼**2,900**× (GPT-2), cross-architecture. **P1:** FPL↔accuracy $\rho = \mathbf{-0.84}$ (p=0.002). **P4:** misclassified inputs sparser, p=$\mathbf{8.2\times10^{-6}}$. **P3:** bias **−70.8%** at $+1.9\%$ perplexity.

4. INTERPRETATION & CONCLUSIONS

Scorecard (6 problems):

- **P6 Solved** — memorization detector, no held-out data
- **P4 Supported** — failure flagged pre-output
- **P1 Deepened** — first topological robustness measure
- **P3 Partial** — works on separable bias; fails when bias is redundantly encoded (COMPAS, toxicity)
- **P5 Negative** — topology does not predict shift-fragility
- **P7 Refuted** — 3-phase structure not universal (ViT)

Innovation. First framework to read learning as topological evolution across training; first validation-free memorization test; first topological concept-robustness measure.

Limitations. ≤124M scale; binary bias groups; maxdim 1; P5 open.

Future. 7B-scale memorization test; per-input FPL for P5; erasure during pre-training; topology-guided curriculum learning.

The honest scope — two strong positives, one new measure, one scoped intervention, two clean negatives — is the result.